Coefficient (in front) and subscript (small, inside the formula)

✓ We CAN change the coefficient

$$2H_2O$$

↑ **coefficient: 2 molecules** ↑ **subscript: 2 H in each one**



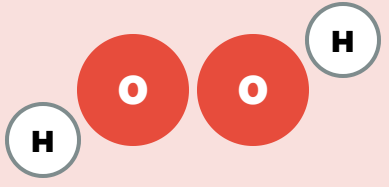
H: **2** · **2** = **4 atoms** · **O**: **2** · 1 = **2 atoms**

The coefficient **multiplies** all the atoms of the formula. It is still **water**.

✗ We can NEVER change the subscripts

$$H_2O_2$$

↑ **we changed a subscript!**



H_2O_2 is **hydrogen peroxide** (the antiseptic for wounds), a **different substance**.

Changing a subscript changes the substance!

To balance, we only write **whole numbers** (2, 3, 4…) **in front** of the formulas: the **stoichiometric coefficients**. A coefficient of 1 is not written.